

FitTrack Development Plan

Full-Stack Team Breakdown and Delivery Guide

Project: FitTrack SaaS Fitness Platform

Date: March 27, 2026

Team Size: 6 Members

Frontend + PHP Backend + MySQL Database

Contents

1	Overall Architecture	2
1.1	Target Architecture	2
1.1.1	Frontend	2
1.1.2	Backend	2
1.1.3	Database	2
2	Team Plan by Member	2
2.1	Member 1: Authentication System Developer	2
2.1.1	Main Goal	2
2.1.2	Pages to Build	2
2.1.3	HTML Responsibilities	2
2.1.4	CSS Responsibilities	3
2.1.5	JavaScript Responsibilities	3
2.1.6	PHP Responsibilities	3
2.1.7	MySQL Table	4
2.2	Member 2: Dashboard Developer	4
2.2.1	Page	4
2.2.2	Main Goal	4
2.2.3	HTML Responsibilities	4
2.2.4	CSS Responsibilities	4
2.2.5	JavaScript Responsibilities	4
2.2.6	PHP Responsibilities	5
2.2.7	MySQL Table	5
2.3	Member 3: Diary System Developer	5
2.3.1	Page	5
2.3.2	Main Goal	5
2.3.3	HTML Responsibilities	5
2.3.4	CSS Responsibilities	6
2.3.5	JavaScript Responsibilities	6
2.3.6	PHP Responsibilities	6
2.3.7	MySQL Tables	7
2.4	Member 4: Food Database Developer	7
2.4.1	Page	7
2.4.2	Main Goal	7
2.4.3	HTML Responsibilities	7
2.4.4	CSS Responsibilities	8
2.4.5	JavaScript Responsibilities	8
2.4.6	PHP Responsibilities	8
2.4.7	MySQL Table	8
2.5	Member 5: Exercise System Developer	8
2.5.1	Page	9
2.5.2	Main Goal	9
2.5.3	HTML Responsibilities	9
2.5.4	CSS Responsibilities	9
2.5.5	JavaScript Responsibilities	9

2.5.6	PHP Responsibilities	9
2.5.7	MySQL Table	10
2.6	Member 6: Goals and Progress Developer	10
2.6.1	Pages	10
2.6.2	Goals Page	10
2.6.3	Progress Page	11
3	Final Database Overview	11
4	CSS Palette from FitTrack Design System	11
4.1	Primary and Accent Colors	11
4.2	Neutral Colors	12
5	Delivery and Integration Notes	12
5.1	Branching Strategy	12
5.2	Recommended Merge Order	13
5.3	Minimum Acceptance Checklist	13
6	Final Result	13

1. Overall Architecture

1.1 Target Architecture

The project evolves from a static frontend into a complete full-stack web application.

1.1.1 Frontend

- HTML for page structure.
- CSS for design system and visual consistency.
- JavaScript for interactions, animations, and UI behavior.

1.1.2 Backend

- PHP for server-side logic.
- Authentication and session management.
- CRUD operations for app data.

1.1.3 Database

- MySQL for persistent storage.
- Stores users, meals, exercises, goals, progress, and daily metrics.

2. Team Plan by Member

2.1 Member 1: Authentication System Developer

Branch: auth-system

2.1.1 Main Goal

Implement user registration, login, logout, and session management.

2.1.2 Pages to Build

- /auth/login.php
- /auth/register.php
- /auth/logout.php

2.1.3 HTML Responsibilities

Create form UIs for registration and login.

Register Form Fields

- Name
- Email
- Password

- Confirm Password
- Register Button

Login Form Fields

- Email
- Password
- Login Button

2.1.4 CSS Responsibilities

Use the existing FitTrack classes for visual consistency:

- .form-group
- .form-input
- .btn
- .card

2.1.5 JavaScript Responsibilities

Implement client-side validation:

- Email format validation
- Password minimum length
- Password confirmation matching

2.1.6 PHP Responsibilities

Register Logic

1. Receive form data.
2. Validate input.
3. Hash password.
4. Insert user into database.

Login Logic

1. Find user by email.
2. Verify password hash.
3. Create authenticated session.

Session Example (Concept)

```
session_start();  
$_SESSION['user_id'] = $user['id'];
```

2.1.7 MySQL Table

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  email VARCHAR(150) UNIQUE,  
  password VARCHAR(255),  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

2.2 Member 2: Dashboard Developer

Branch: dashboard-backend

2.2.1 Page

- index.php

2.2.2 Main Goal

Convert the static dashboard into dynamic user-specific data.

2.2.3 HTML Responsibilities

Reuse current dashboard sections:

- Stats cards
- Calorie ring
- Meals summary
- Exercise summary
- Water tracker

2.2.4 CSS Responsibilities

Continue using current components:

- .stat-card
- .progress-bar
- .calorie-ring

No major styling changes required.

2.2.5 JavaScript Responsibilities

Keep and adapt current interactions:

- Count-up animations
- Progress bar animations
- Calorie ring animation
- Fetch dynamic values if needed

2.2.6 PHP Responsibilities

Fetch daily statistics for the logged-in user.

Examples:

- Calories consumed today
- Calories burned today
- Water intake today
- Exercise count today

Sample Query

```
SELECT SUM(calories)
FROM meals
WHERE user_id = ? AND date = CURDATE();
```

2.2.7 MySQL Table

```
CREATE TABLE daily_stats (
  id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT,
  date DATE,
  water_glasses INT,
  calories_consumed INT
);
```

2.3 Member 3: Diary System Developer

Branch: diary-system

2.3.1 Page

- diary.php

2.3.2 Main Goal

Allow users to log meals and exercises per day.

2.3.3 HTML Responsibilities

Build and connect these sections:

- Date navigation
- Food log
- Exercise log
- Water tracker
- Notes

Forms

- Add meal modal: Food name, calories, protein, carbs, fat, meal type.
- Add exercise modal: Exercise name, duration, calories burned.

2.3.4 CSS Responsibilities

Use existing styles:

- .diary-header
- .diary-entry
- .modal
- .progress-bar

2.3.5 JavaScript Responsibilities

Water Tracker

- Toggle glass filled state on click.
- Update water counter.

Date Navigation

- Previous day button.
- Next day button.
- Refresh displayed entries by selected date.

2.3.6 PHP Responsibilities

Implement diary CRUD operations:

- Add meal
- Fetch meals
- Delete meal
- Add exercise
- Fetch exercises
- Delete exercise

Sample CRUD Queries

```
INSERT INTO meals (...);  
SELECT * FROM meals WHERE user_id = ? AND date = CURDATE();  
DELETE FROM meals WHERE id = ?;
```


2.3.7 MySQL Tables

Meals

```
CREATE TABLE meals (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT,  
  food_name VARCHAR(100),  
  meal_type VARCHAR(50),  
  calories INT,  
  protein INT,  
  carbs INT,  
  fat INT,  
  date DATE  
);
```

Exercise Logs

```
CREATE TABLE exercise_logs (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT,  
  exercise_name VARCHAR(100),  
  duration INT,  
  calories_burned INT,  
  date DATE  
);
```

2.4 Member 4: Food Database Developer

Branch: food-database

2.4.1 Page

- food.php

2.4.2 Main Goal

Create a searchable, filterable food database.

2.4.3 HTML Responsibilities

- Food card grid
- Search bar
- Filter chips

Each food card should show:

- Food name
- Serving
- Calories
- Protein

- Carbs
- Fat

2.4.4 CSS Responsibilities

Use existing classes:

- .food-card
- .food-grid
- .food-filter-chip

2.4.5 JavaScript Responsibilities

- Search by food name.
- Category filter:
 - Fruits
 - Vegetables
 - Protein
 - Grains
 - Dairy
 - Snacks

2.4.6 PHP Responsibilities

Query foods from database by search and filters.

Example Query

```
SELECT * FROM foods
WHERE name LIKE '%search%';
```

2.4.7 MySQL Table

```
CREATE TABLE foods (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  category VARCHAR(50),
  calories INT,
  protein INT,
  carbs INT,
  fat INT
);
```

2.5 Member 5: Exercise System Developer

Branch: exercise-system

2.5.1 Page

- exercise.php

2.5.2 Main Goal

Manage exercise categories and user exercise logs.

2.5.3 HTML Responsibilities

Build and connect:

- Exercise categories section
- Recent exercises section
- Add exercise modal

Required categories:

- Cardio
- Strength
- Flexibility
- Sports

2.5.4 CSS Responsibilities

Use existing styles:

- .exercise-category-card
- .exercise-list-item

2.5.5 JavaScript Responsibilities

- Open add exercise modal.
- Submit form.
- Update recent exercise list.

2.5.6 PHP Responsibilities

- Insert exercise logs.
- Fetch recent exercises.

Example Query

```
INSERT INTO exercise_logs (...);
```

2.5.7 MySQL Table

```
CREATE TABLE exercises (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  category VARCHAR(50),  
  calories_per_min INT  
);
```

2.6 Member 6: Goals and Progress Developer

Branch: goals-progress-system

2.6.1 Pages

- goals.php
- progress.php

2.6.2 Goals Page

HTML Responsibilities

- Target weight form
- Daily calorie target form
- Weekly workout target form

JavaScript Responsibilities

- Form validation
- Success toast feedback

PHP Responsibilities

- Save and update goals by user.

Example Query

```
UPDATE goals SET daily_calories = ?;
```

MySQL Table

```
CREATE TABLE goals (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT,  
  target_weight INT,  
  daily_calories INT,  
  weekly_workouts INT  
);
```

2.6.3 Progress Page

HTML Responsibilities

- Weight chart
- Calorie chart
- Monthly goals summary

JavaScript Responsibilities

- Render charts (for example with Chart.js).

PHP Responsibilities

- Fetch historical tracking data.

Example Query

```
SELECT weight, date
FROM weight_logs
WHERE user_id = ?;
```

MySQL Table

```
CREATE TABLE weight_logs (
  id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT,
  weight FLOAT,
  date DATE
);
```

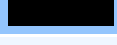
3. Final Database Overview

Table	Purpose
users	User accounts and authentication
foods	Master food catalog
meals	Daily meal logs per user
exercises	Master exercise catalog
exercise_logs	User exercise entries
goals	User goal settings
weight_logs	Historical weight tracking
daily_stats	Daily aggregates (water, calories)

4. CSS Palette from FitTrack Design System

This section reproduces the exact color tokens currently defined in the CSS file.

4.1 Primary and Accent Colors

Token	Hex	Swatch
–primary	#3B82F6	
–primary-dark	#2563EB	
–primary-light	#93C5FD	
–primary-bg	#EFF6FF	
–accent-green	#10B981	
–accent-green-bg	#ECFDF5	
–accent-red	#EF4444	
–accent-red-bg	#FEE2E2	
–accent-orange	#F59E0B	
–accent-orange-bg	#FFFBEA	
–accent-purple	#8B5CF6	
–accent-purple-bg	#F5F3FF	
–accent-teal	#14B8A6	
–accent-pink	#EC4899	

4.2 Neutral Colors

Token	Hex	Swatch
–white	#FFFFFF	
–gray-50	#F9FAFB	
–gray-100	#F3F4F6	
–gray-200	#E5E7EB	
–gray-300	#D1D5DB	
–gray-400	#9CA3AF	
–gray-500	#6B7280	
–gray-600	#4B5563	
–gray-700	#374151	
–gray-800	#1F2937	
–gray-900	#111827	

5. Delivery and Integration Notes

5.1 Branching Strategy

- Each developer works in their dedicated feature branch.
- Open pull requests against a shared integration branch.
- Resolve merge conflicts before release branch cut.

5.2 Recommended Merge Order

1. auth-system
2. dashboard-backend
3. diary-system
4. food-database
5. exercise-system
6. goals-progress-system

5.3 Minimum Acceptance Checklist

- User can register, login, logout.
- Dashboard shows dynamic daily values.
- Diary supports add, read, and delete for meals and exercises.
- Food page supports search and category filtering.
- Exercise page supports adding logs and listing reents.
- Goals save correctly and progress data renders in charts.
- All MySQL tables are created and linked by user_id where required.

6. Final Result

When complete, FitTrack becomes a full SaaS fitness platform with:

- User accounts and secure authentication.
- Daily food tracking and nutrition logging.
- Exercise logging and category management.
- Goal management (weight, calories, weekly workouts).
- Progress visualization through charts.
- Searchable food database.

End of Document